

WHY PYBIO-10

From the bench to the **code.**

Spreadsheets break when the data gets big. In 10 evenings you learn to clean, analyse, visualise and model your own biological data in Python, the language used across modern genomics, bioinformatics and lab analytics.

```
# Day 7: your first sequence scan
from Bio import SeqIO
from Bio.SeqUtils import gc_fraction
for r in SeqIO.parse("genes.fasta", "fasta"):
    print(r.id, gc_fraction(r.seq))
```

By Day 10 you will be able to

- Write clean, reusable Python in Jupyter and Google Colab
- Turn messy lab and field data into tidy, analysis-ready tables
- Run t-tests, ANOVA and regression, and make publication-ready figures
- Read FASTA and GenBank files, run BLAST, and handle gene-expression data
- Build and evaluate your first machine-learning model for biology



AVYAGRAHA RESEARCH AND ANALYTICS LLP
PRESENTS

REGISTRATIONS OPEN

PyBio-10

Python for Life Sciences

10-DAY HANDS-ON TRAINING PROGRAMME

DATES

16 – 26 Nov 2026

Sunday 22 Nov holiday

TIMING

6:30 – 8:30 PM IST

2 hours per evening

MODE

Online · Live

Hands-on, no prior coding

FEE (INCL. GST)

₹1,499 / ₹1,999

Students / Professionals

Python

NumPy

Pandas

Matplotlib

Seaborn

SciPy

Biopython

scikit-learn

WHO CAN REGISTER

Built for life scientists, not programmers.

- UG, PG and PhD students in the life sciences
- Faculty, scientists and research associates
- Biotechnology, Microbiology, Biochemistry, Genetics, Agriculture, Pharmacy and Medical Sciences
- Pharma, agri-biotech and healthcare professionals
- Anyone moving from Excel or SPSS to code-based analysis

No prior programming experience needed. A laptop with internet access and a Google account (for Colab) is all you need.

**Start where you are.
Code your own data.**

ENROLL NOW



aviqlabs.com/pybio10

Students

UG / PG / PhD students and research scholars

₹1,499

Professionals

Faculty, scientists, industry and others

₹1,999

All fees inclusive of GST · Limited seats

- 1 Scan the QR or visit aviqlabs.com/pybio10
- 2 Tap **Pay now**: your UPI app opens with the fee filled in
- 3 Note the **UPI transaction ID** after paying
- 4 Enter it, **upload the payment screenshot** and submit

COURSE STRUCTURE

10 days. 10 skills.

Every session is live and hands-on: you write and run the code yourself, on biological datasets.

Module 1 • Python Foundry Days 1–2

D01 Python foundations
Jupyter, Google Colab, core syntax
Mon 16 Nov

D02 Thinking in code
Data types, loops, conditions, functions
Tue 17 Nov

Module 2 • Scientific Data Toolkit Days 3–5

D03 NumPy for science
Arrays and fast numerical computing
Wed 18 Nov

D04 Pandas for lab data
Import, clean, merge and reshape datasets
Thu 19 Nov

D05 Publication-ready plots
Matplotlib and Seaborn
Fri 20 Nov

Module 3 • Stats and Sequences Days 6–7

D06 Biostatistics in Python
t-tests, ANOVA, regression with SciPy
Sat 21 Nov

D07 Biopython essentials
Sequences, FASTA, GenBank, BLAST
Mon 23 Nov

Module 4 • Omics and ML Days 8–9

D08 Omics data handling
Gene expression and NGS file formats
Tue 24 Nov

D09 Machine learning for biology
Classification and clustering with scikit-learn
Wed 25 Nov

Module 5 • Capstone Day 10

D10 Capstone and certification
Solve a real research problem end to end
Thu 26 Nov

Sunday 22 November

HOLIDAY • NO SESSION

Every session: 6:30 – 8:30 PM IST

TOOLS YOU WILL USE

Python 3

Jupyter

Google Colab

NumPy

Pandas

Matplotlib

Seaborn

SciPy

statsmodels

Biopython

scikit-learn

ORGANIZER AND RESOURCE PERSONS

Learn from practitioners.



ORGANIZER AND SPEAKER

Dr. Avinash Gopa

Founder & CEO, Avyagraha Research & Analytics LLP · Ex-ICMR Project Scientist-II (ICMR-NIOH, Ahmedabad) · Ph.D. (Agricultural Statistics), IARI, New Delhi



GUEST SPEAKER

Mr. Rajesh A

CEO & Tech Member, Avinya Techsphere, Bangalore



GUEST SPEAKER

Mr. Chandan G Y

Data Analyst, Bangalore

How sessions run. Each evening opens with a short concept walk-through, then you code along on a shared notebook, finishing with an exercise and open Q&A.

KEY TAKEAWAYS

You walk away with



Code notebooks

Every day's notebook, ready to reuse on your own research data



Recordings for 3 months

Revisit any session at your own pace



Capstone project

An end-to-end analysis you can show on your CV



E-certificate

Issued on completion of the programme



Real biological datasets

Sequence, expression, lab and field data from published studies



Live doubt clearing

Direct interaction with the trainers in every session

ABOUT AVYAGRAHA



A DPIIT-recognised deep-tech startup, incubated at AIC-UAS-B, GKVK, Bengaluru. We provide data-driven solutions, training programmes and research collaborations.

- **Training:** statistics, R, Python, AI/ML and Generative AI for institutions and professionals
- **Research consultancy:** study design, data analysis and statistical modelling
- **Partnerships:** MoUs with universities and R&D institutes

Contact us

☎ +91 83105 79116 · +91 94803 63456

✉ avinashstat@aviqlabs.com

🌐 www.aviqlabs.com

PyBio-10: Python for Life Sciences

10-Day Hands-On Training Programme

16 Nov 2026 – 26 Nov 2026 · 6:30 PM to 8:30 PM IST · Online (live) · Sunday holiday

Organized by Avyagraha Research and Analytics LLP, Ramasagara, Ballari, Karnataka

www.aviqlabs.com · avinashstat@aviqlabs.com · +91 83105 79116 · +91 94803 63456

Date	Topics	Resource person
Module 1 • Python Foundry (Days 1–2)		
16/11/2026 Monday	D01 Inauguration • Python foundations and setup Programme overview and expectations. Installing Python, Jupyter Notebook and Google Colab; notebooks, cells and markdown; variables, operators, basic input/output; your first script on a small lab dataset.	Dr. Avinash Gopa Founder & CEO, Avyagraha & Mr. Chandan G Y Data Analyst, Bangalore
17/11/2026 Tuesday	D02 Thinking in code Data types (int, float, str, bool); lists, tuples, dictionaries and sets; if-elif-else; for and while loops; writing functions with arguments and return values. Exercise: codon counting and assay unit conversions.	Mr. Chandan G Y Data Analyst, Bangalore
Module 2 • Scientific Data Toolkit (Days 3–5)		
18/11/2026 Wednesday	D03 NumPy for science Creating arrays; indexing and slicing; broadcasting and vectorised maths; aggregation along axes; random numbers and simple simulation. Exercise: processing a plate-reader absorbance matrix.	Mr. Rajesh A CEO & Tech Member, Avinya Techsphere, Bangalore
19/11/2026 Thursday	D04 Pandas for lab data Series and DataFrames; reading CSV and Excel files; inspecting, filtering and sorting; handling missing values; groupby, merge and pivot/melt to build tidy experimental datasets.	Mr. Chandan G Y Data Analyst, Bangalore
20/11/2026 Friday	D05 Publication-ready plots Matplotlib figure anatomy; Seaborn bar plots with error bars, box and violin plots, scatter with regression line and heatmaps; multi-panel figures; exporting journal-ready figures (300 dpi TIFF/PDF).	Mr. Rajesh A CEO & Tech Member, Avinya Techsphere, Bangalore
Module 3 • Stats and Sequences (Days 6–7)		
21/11/2026 Saturday	D06 Biostatistics in Python Descriptive statistics; checking normality; t-tests and chi-square tests; one-way and two-way ANOVA with Tukey post-hoc; correlation and linear regression using SciPy and statsmodels.	Dr. Avinash Gopa Founder & CEO, Avyagraha
22/11/2026 Sunday	HOLIDAY	

Date	Topics	Resource person
23/11/2026 Monday	D07 Biopython essentials Seq objects; transcription, translation and GC content; parsing FASTA and GenBank with SeqIO; fetching records from NCBI with Entrez; running and parsing BLAST; pairwise alignment.	Mr. Rajesh A CEO & Tech Member, Avinya Techsphere, Bangalore
Module 4 • Omics and Machine Learning (Days 8–9)		
24/11/2026 Tuesday	D08 Omics data handling Gene-expression count matrices; normalisation and log transformation; fold change, p-values and FDR correction; volcano plots and clustered heatmaps; overview of FASTQ, SAM/BAM and VCF formats.	Dr. Avinash Gopa Founder & CEO, Avyagraha
25/11/2026 Wednesday	D09 Machine learning for biology The ML workflow in scikit-learn; train/test split and cross-validation; classification with logistic regression and random forest; confusion matrix and ROC curve; k-means and hierarchical clustering; PCA on expression data.	Mr. Chandan G Y Data Analyst, Bangalore
Module 5 • Capstone (Day 10)		
26/11/2026 Thursday	D10 Capstone, certification and valedictory End-to-end capstone on a real research dataset: clean, analyse, visualise, model and report. Short presentations with feedback; sharing notebooks on GitHub; valedictory session and e-certificates.	Dr. Avinash Gopa Founder & CEO, Avyagraha

Benefits

- Live, hands-on sessions: you write and run every piece of code yourself, on biological datasets.
- Daily code notebooks to reuse directly in your own research.
- Class recordings with **3-month access**.
- One end-to-end **capstone project** for your CV or portfolio.
- **E-certificate** on completion of the programme.
- Live doubt-clearing and direct interaction with the trainers.
- No prior programming experience needed; everything runs in the browser through Google Colab.

Programme fees

Category	Fee (inclusive of GST)
Students: UG / PG / PhD students and research scholars	Rs. 1,499 /-
Professionals: faculty, scientists, research associates, industry and all other participants	Rs. 1,999 /-

How to register

1. Scan the QR code or visit aviqlabs.com/pybio10.
2. Tap **Pay now**. Google Pay, PhonePe, Paytm or any UPI app opens with the fee filled in. You can also pay to UPI ID **9480363456@okbizaxis** (Avyagraha Research and Analytics LLP).
3. Note the **UPI transaction ID** shown after payment.
4. Enter the transaction ID, **upload the payment screenshot** and submit. You receive a registration ID and a confirmation email.
5. Joining details will be shared by email and WhatsApp before the programme begins.



aviqlabs.com/pybio10

Requirements

- A laptop or desktop with a stable internet connection.
- A Google account, for Google Colab notebooks.

* * * * *